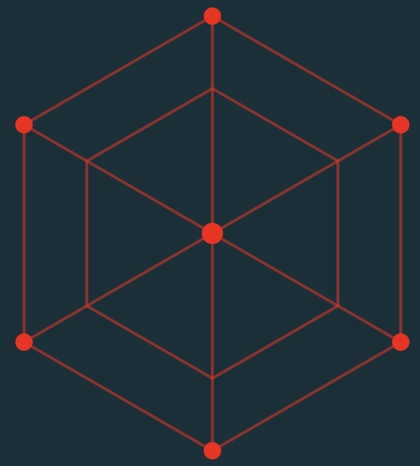
PINECREST HEALTH · CARE COORDINATION



Pinecrest Care Coordinator

Ranking 30-day readmission risk and turning it into an approved care action — one governed data journey on Databricks, from raw files to the app.

Narasimha Kamathardi · Fictional hospital system, synthetic data

November 2026

THE PROBLEM

Coordinators can only call a fraction of discharged patients — today they don't know who to call first.

HOSPITALS

3

Pinecrest Main, North, Valley

DISCHARGES IN DATASET

6,013

after data-quality checks

30-DAY READMISSION RATE

14.7%

about 1 in 7 come back

PENALTY EXPOSURE

HRRP

CMS readmission penalties

SO WHAT

Every avoidable readmission costs a second stay plus Medicare penalty risk — and coordinators work the list without knowing who matters most.

THE REAL QUESTION

"Which of today's discharges is most likely to come back, why — and what do we do first?"

HOW IT'S BUILT

Six stages, one journey — every stage reads what the previous one produced.

01 →

Lakeflow

Auto Loader ingests raw CSV/JSON from a Unity Catalog volume into bronze → silver → gold, with data-quality expectations.

02 →

Unity Catalog

PHI tags, column masks on name and phone, business comments and automatic lineage.

03 →

ML

Gradient-boosted readmission model, tracked in MLflow, registered in UC as *champion*, scores every patient.

04 →

Lakebase

Risk worklist and notes synced from UC into Postgres, plus a writable `care_actions` table.

05 →

Genie

Natural-language questions over the same governed gold tables.

06

Databricks App

Worklist, AI explanation (Claude via a serving endpoint), approve → write back to Lakebase.

One platform, one permission model — raw file, model score, served row and approved action are all governed by Unity Catalog.

DATA & GOVERNANCE

Bad data is stopped at ingest, and patient identity never leaks.

LAKEFLOW PIPELINE

2,000 patients · **6,038** raw discharges · **3,000** coordinator notes

25 invalid rows dropped by the `valid_length_of_stay` expectation

6,013 clean discharges in gold, ready for the model

Serverless, declarative, incremental (Auto Loader)

UNITY CATALOG

Name, phone and ZIP tagged `phi`

Name and phone **masked** for anyone outside the care team

Business comments on the risk table (also improves Genie)

Lineage captured bronze → silver → gold automatically

All data generated synthetically with a fixed seed — no real patients, no customer data.

Evidence: evidence/01_lakeflow · evidence/02_unity_catalog

THE MODEL

The top 10% of patients the model flags are readmitted 3.3× as often as average.

TEST AUC	AVERAGE PRECISION	TOP-DECILE LIFT	REGISTERED
0.779	**0.434**	**3.34×**	**v1**
held-out 25% of discharges	vs 0.147 base rate	where coordinators should call first	@champion in Unity Catalog

OUTCOME

2,000 patients scored → **253 High**, **375 Medium**, **1,372 Low** risk — each with a next-best action, e.g. "*Schedule follow-up visit within 7 days*".

WHERE THE RISK SITS

Patients who left against medical advice average **0.29** predicted risk vs **0.13** for those discharged home (from Genie, slide 6).

Evidence: evidence/03_ml_training_run_output.md

GENIE

Ask it in plain English — Genie writes the SQL on the same governed tables.

"How many high-risk patients does each facility have?"	Main **91** · North **89** · Valley **73**
"Which facility had the highest COPD readmission rate?"	Pinecrest North — **20.8%** (58 of 279)
"Average predicted risk by discharge disposition?"	Left AMA **0.29** vs Home **0.13** — more than 2× higher

Evidence: evidence/05_genie_conversations.md

ONE PATIENT, END TO END

Patient PC-01684: CHF, skilled nursing, no follow-up booked

1 **Lakeflow**
Raw file → bronze → silver → gold

2 **Unity Catalog**
Name and phone masked

3 **ML**
Risk **0.91** → **High**: "Schedule follow-up within 7 days"

4 **Genie**
#1 answer to "high-risk patients with no follow-up"

5 **Lakebase**
Served to the app in **~4 ms**

6 **App**
AI explanation → coordinator approves → written to `care_actions`

Human approval before anything is written — every action records the approver and timestamp in Lakebase.

Evidence: evidence/06_app_execution.md

VALUE & NEXT STEP

One point of readmission rate is worth ~$3M a year at a 20,000-discharge system.

DISCHARGES / YR	−1 POINT OF RATE	× COST EACH	= AVOIDED COST
20,000	−200	$15K	$3M
mid-size health system	readmissions avoided	second stay	per year, per point

Illustrative industry assumptions, not from the synthetic data.

01 · PILOT
90 days on CHF and COPD discharges

02 · MEASURE
Follow-ups booked within 7 days, weekly

03 · IMPROVE
Retrain monthly from the same pipeline